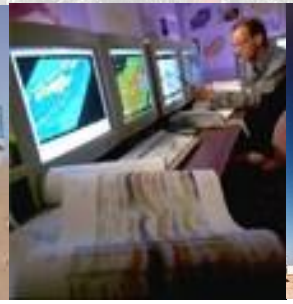
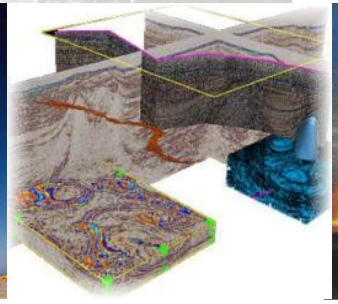
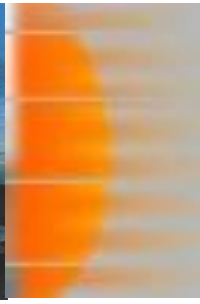
ENG-3020
RENEWABLE ENERGY SOURCES









A REVIEW OF THERMAL SCIENCES

Thermal Sciences

Thermal sciences: The physical sciences that deal with energy and the transfer, transport, and conversion of energy

The analysis of all kinds energy systems requires understanding of

- energy conversion
- transformation between various forms of energy
- ways of defining efficiencies of energy systems

For these requirements, it is essential to review fundamental concepts of

- thermodynamics
- heat transfer
- fluid mechanics
- thermochemistry
- power plants

THERMODYNAMICS

Thermodynamics: The science of energy.

- Conservation of energy principle: During a process, energy can change from one form to another but the total amount of energy remains constant.

  Energy cannot be created or destroyed.

- The first law of thermodynamics: It is the principle expression of the conservation of energy.

- The second law of thermodynamics: It asserts that energy has quality as well as quantity, and actual processes occur in the direction of decreasing quality of energy.

Heat and other forms of energy

Energy can exist or can be obtained in numerous forms such as

- thermal
- mechanical
- kinetic
- potential
- electrical
- magnetic
- chemical
- Nuclear

Their sum constitutes the total energy of a system.

Internal energy (u): The sum of all microscopic forms of energy

FUNDAMENTALS

Dimensions and Units

- Any physical quantity can be characterized by its **dimensions**.
- The magnitudes assigned to the dimensions are called **units**.
- Some basic dimensions such as mass *m*, length *L*, time *t*, and temperature *T* are selected as **primary** or **fundamental dimensions**, while others such as velocity *V*, energy *E*, and volume *v* are expressed in terms of the primary dimensions and are called **secondary dimensions**, or **derived dimensions**.

- The **SI** **is a simple and logical system** based on a **decimal relationship between the various units**, and it is being used for scientific and engineering work in most of the industrialized nations.

- The **English system**, however, has **no apparent systematic numerical base**, and various units in this system are related to each other rather arbitrarily (12 in=1 ft, 1 mile=5280 ft, etc.) which makes it confusing and difficult to learn.

The seven fundamental (or primary) dimensions and their units in SI

Dimension	Unit
Length	meter (m)
Mass	kilogram (kg)
Time	second (s)
Temperature	kelvin (K)
Electric current	ampere (A)
Amount of light	candela (cd)
Amount of matter	mole (mol)

Standard prefixes in SI units

Multiple	Prefix
10^{12}	tera, T
10^{9}	giga, G
10^{6}	mega, M
10^{3}	kilo, k
10^{2}	hecto, h
10^{1}	deka, da
10^{-1}	deci, d
10^{-2}	centi, c
10^{-3}	milli, m
10^{-6}	micro, μ
10^{-9}	nano, n
10^{-12}	pico, p

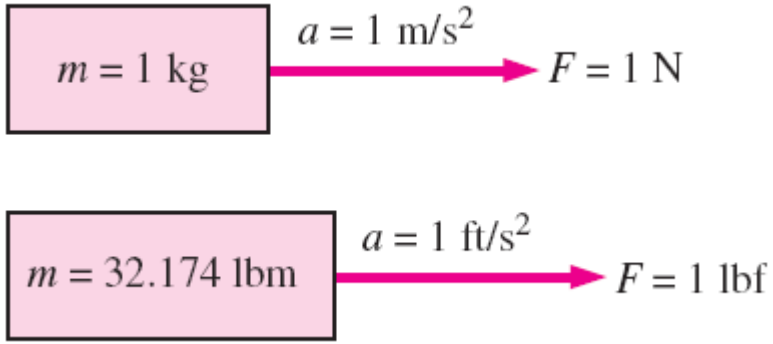
Work and Force

Force = Mass × Acceleration

$$F = ma$$

$$1\ \text{N} = 1\ \text{kg} \cdot \text{m/s}^2$$

$$1\ \text{lbf} = 32.174\ \text{lbm} \cdot \text{ft/s}^2$$



Work = Force× Distance

1 J = 1 N·m

1 cal = 4.1868 J

1 Btu = 1.0551 kJ

Pressure

Pressure: A normal force exerted by a fluid per unit area

$$1 \text{ bar} = 10^5 \text{ Pa} = 0.1 \text{ MPa} = 100 \text{ kPa}$$

$$1 \text{ atm} = 101{,}325 \text{ Pa} = 101.325 \text{ kPa} = 1.01325 \text{ bars}$$

$$1 \text{ kgf/cm}^2 = 9.807 \text{ N/cm}^2 = 9.807 \times 10^4 \text{ N/m}^2 = 9.807 \times 10^4 \text{ Pa}$$

$$= 0.9807 \text{ bar}$$

$$= 0.9679 \text{ atm}$$

$$1 \text{ Pa} = 1 \text{ N/m}^2$$

Pascal's Law

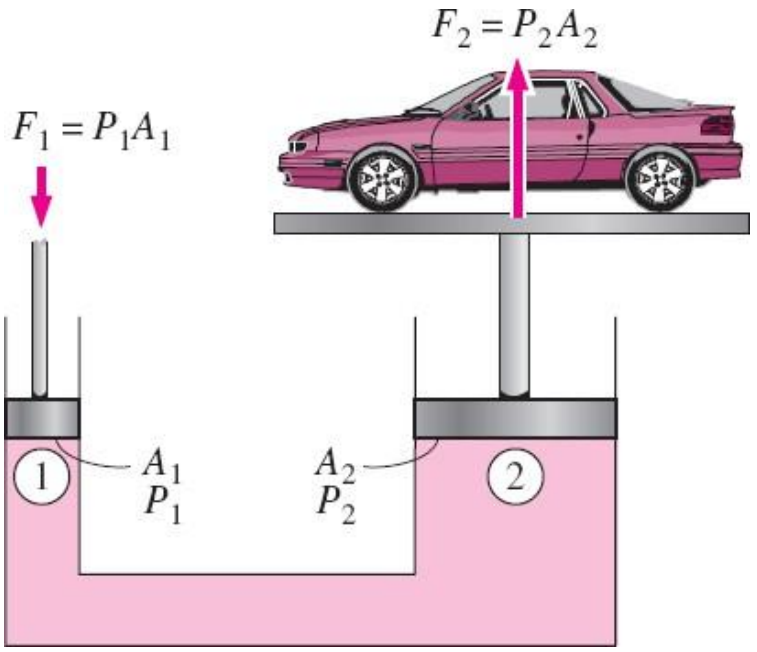
Pascal's law: The pressure applied to a confined fluid increases the pressure throughout by the same amount.

$$P_1 = P_2 \quad \rightarrow \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \rightarrow \quad \frac{F_2}{F_1} = \frac{A_2}{A_1}$$

The area ratio A_2/A_1 is called the *ideal mechanical advantage* of the hydraulic lift.

Using a hydraulic car jack with a piston area ratio of $A_2/A_1 = 10$,

For example, a person can lift a 1000-kg car by applying a force of just 100 kgf (=908 N)



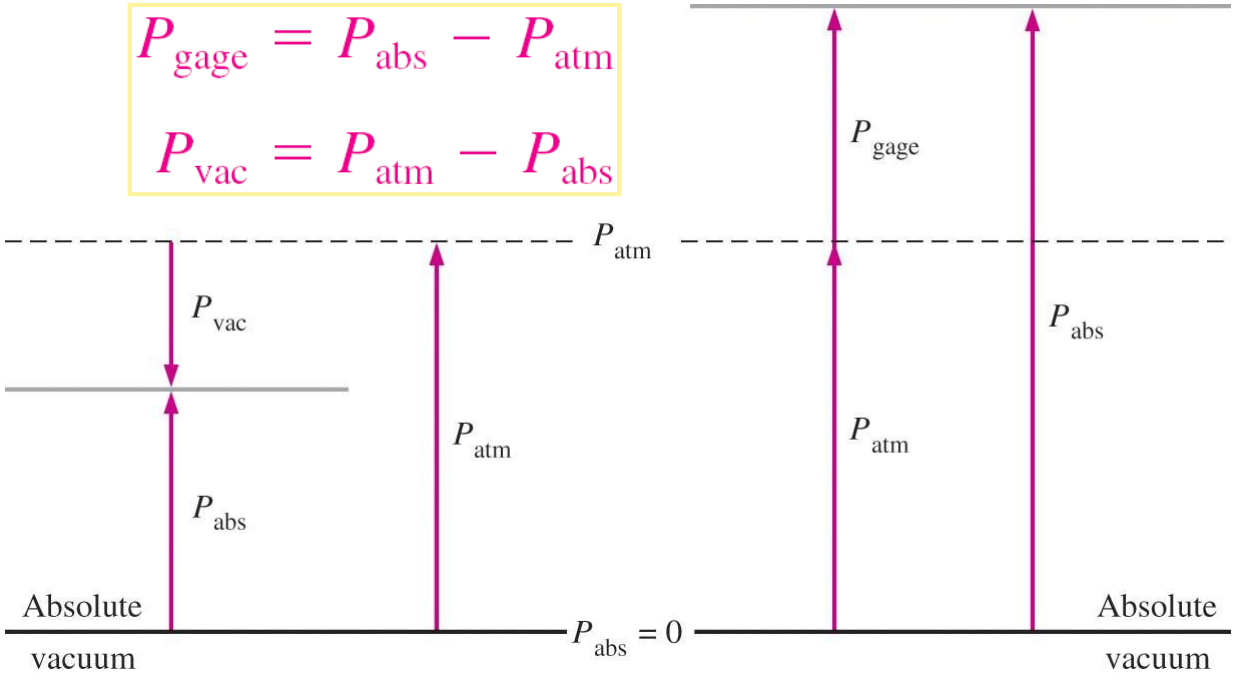
Lifting of a large weight by a small force by the application of Pascal's law.

Pressure

- **Absolute pressure**: The actual pressure at a given position. It is measured relative to absolute vacuum (i.e., absolute zero pressure).
- **Gage pressure**: The difference between the absolute pressure and the local atmospheric pressure. Most pressure-measuring devices are calibrated to read zero in the atmosphere, and so they indicate gage pressure.
- **Vacuum pressures**: Pressures below atmospheric pressure.

$$P_{gage} = P_{abs} - P_{atm}$$

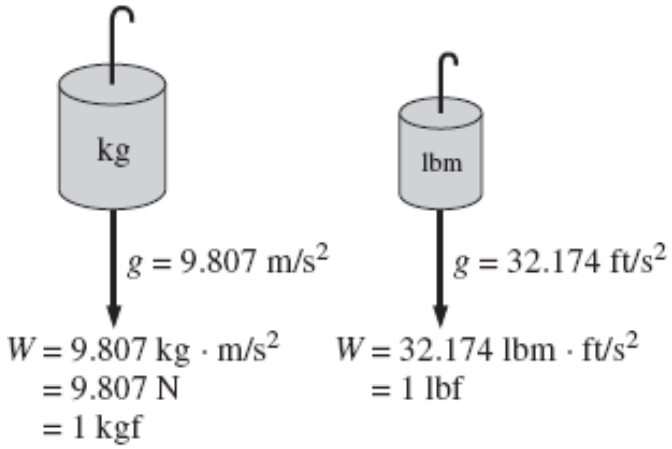
$$P_{vac} = P_{atm} - P_{abs}$$



Mass and Weight

$$W = mg \quad (\text{N})$$

W weight
m mass
g gravitational acceleration



The weight of a unit mass at sea level.



A body weighing 60 kgf on earth will weigh only 10 kgf on the moon.

The relative magnitudes of the force units newton (N), kilogram-force (kgf), and pound-force (lbf).

Density, Specific Volume, Specific Gravity

Density

Density is the mass per unit volume; Specific volume is the volume per unit mass.

$$\rho = \frac{m}{V} \qquad (\text{kg/m}^3)$$

$V = 12\ \text{m}^3$
$m = 3\ \text{kg}$

$\rho = 0.25\ \text{kg/m}^3$

$v = \frac{1}{\rho} = 4\ \text{m}^3/\text{kg}$

Specific volume

$$v = \frac{V}{m} = \frac{1}{\rho}$$

Specific weight: The weight of a unit volume of a substance.

$$\gamma_s = \rho g \qquad (\text{N/m}^3)$$

Specific gravity: The ratio of the density of a substance to the density of some standard substance at a specified temperature (usually water at 4°C).

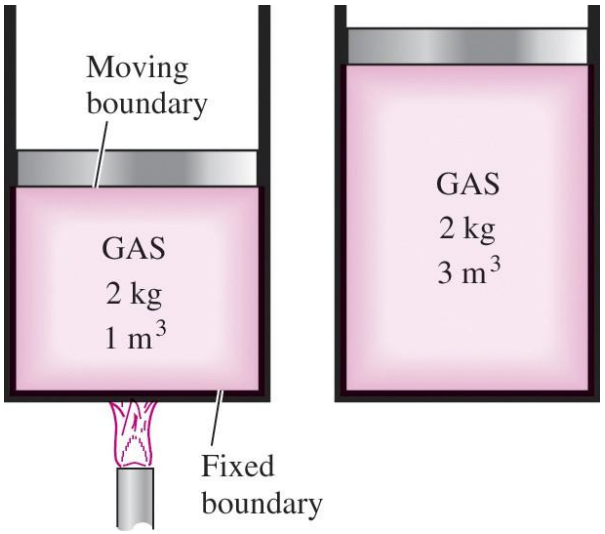
$$\text{SG} = \frac{\rho}{\rho_{H_2O}}$$

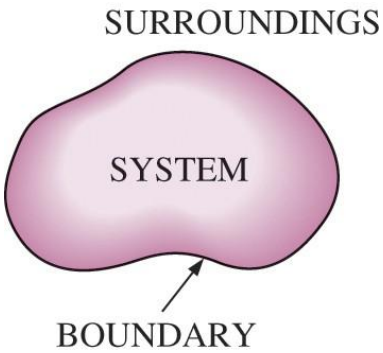
Specific gravities of some substances at 0°C

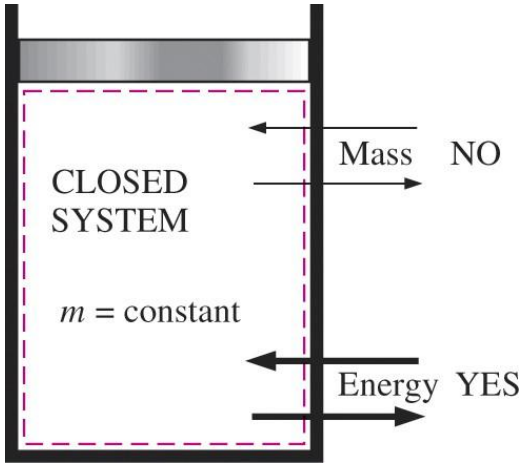
Substance	SG
Water	1.0
Blood	1.05
Seawater	1.025
Gasoline	0.7
Ethyl alcohol	0.79
Mercury	13.6
Wood	0.3–0.9
Gold	19.2
Bones	1.7–2.0
Ice	0.92
Air (at 1 atm)	0.0013

Closed Systems

- **System**: A quantity of matter or a region in space chosen for study.
- **Surroundings**: The mass or region outside the system
- **Boundary**: The real or imaginary surface that separates the system from its surroundings.
- The boundary of a system can be *fixed* or *movable*.
- Systems may be considered to be *closed* or *open*.



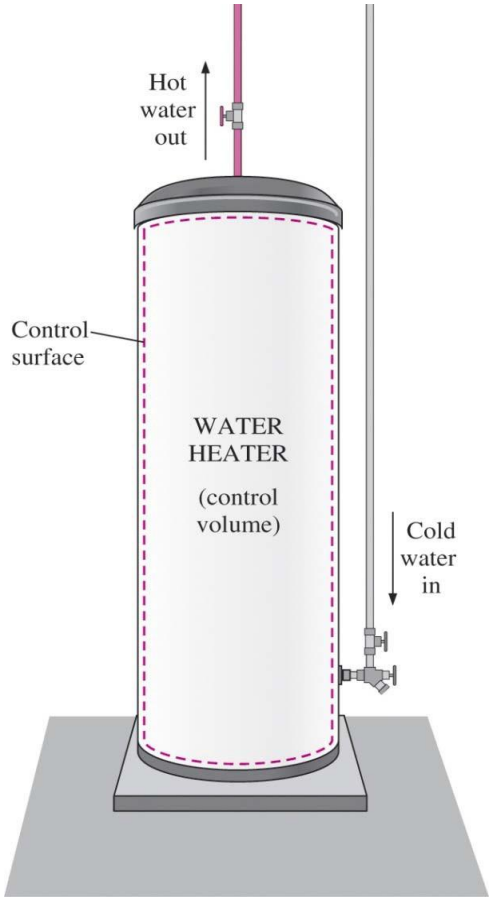




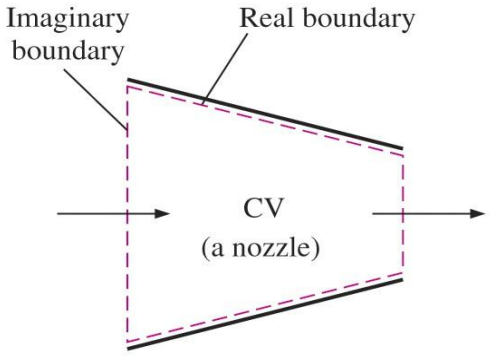
Closed system (Control mass): A fixed Amount of mass, and no mass can cross its boundary.

Open Systems

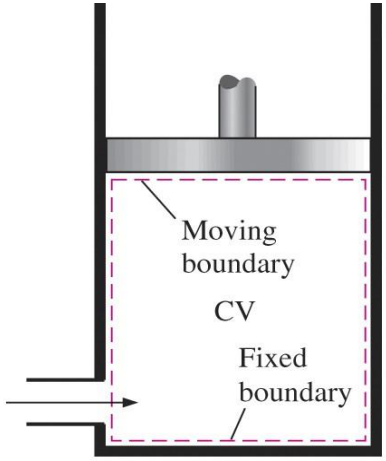
- **Open system (control volume)**: A properly selected region in space.
- It usually encloses a device that involves mass flow such as a compressor, turbine, or nozzle.
- Both mass and energy can cross the boundary of a control volume.
- **Control surface**: The boundaries of a control volume. It can be real or imaginary.



An open system (a control volume) with one inlet and one exit.



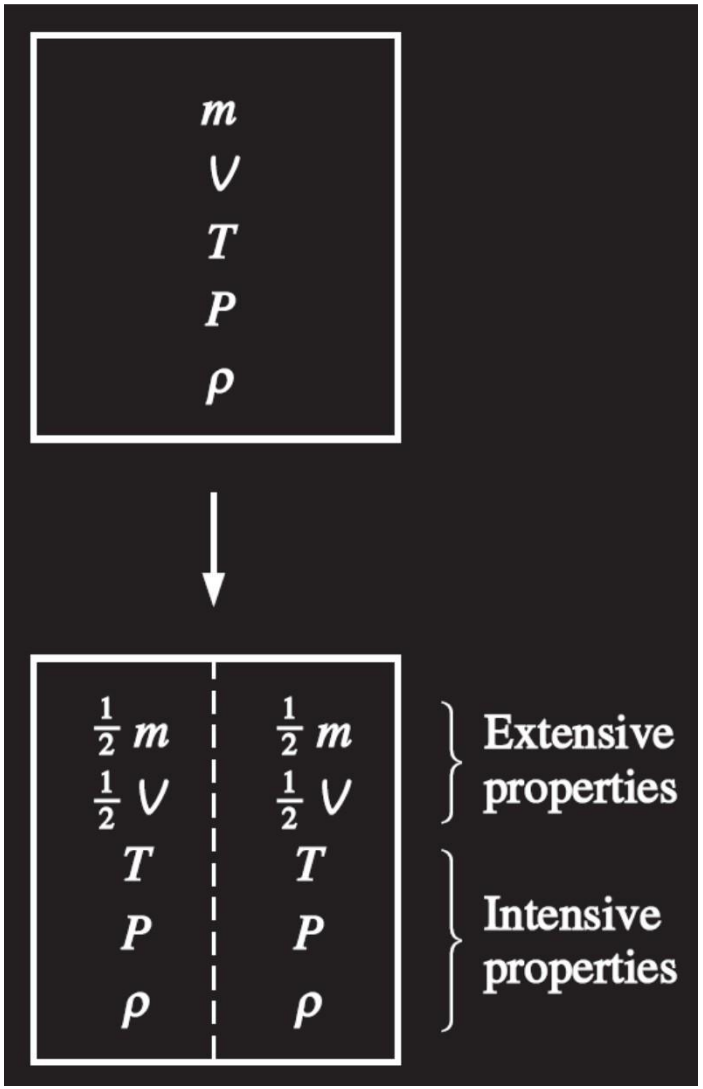
(*a*) A control volume with real and imaginary boundaries



(*b*) A control volume with fixed and moving boundaries

Properties of a system

- **Property:** Any characteristic of a system.
- Some familiar properties are pressure ***P***, temperature ***T***, volume ***V,*** and mass ***m***.
- Properties are considered to be either *intensive* or *extensive*.
- **Intensive properties:** Those that are independent of the mass of a system, such as temperature, pressure, and density.
- **Extensive properties:** Those whose values depend on the size-or extent-of the system.
- **Specific properties:** Extensive properties per unit mass.

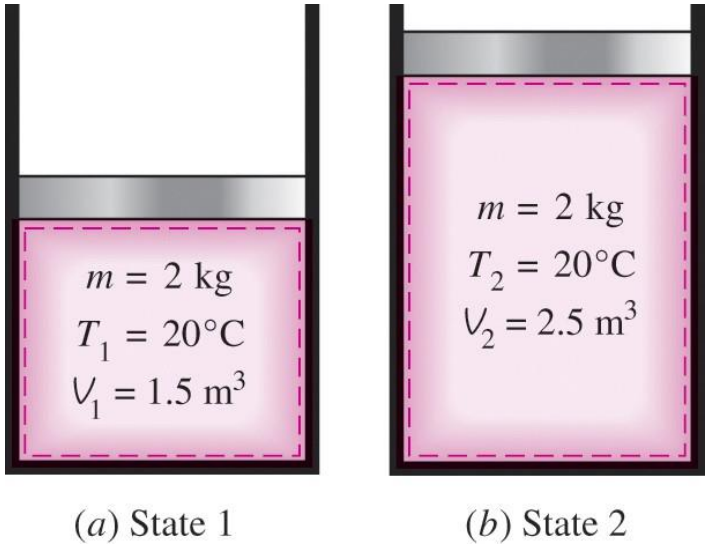


Criterion to differentiate intensive and extensive properties.

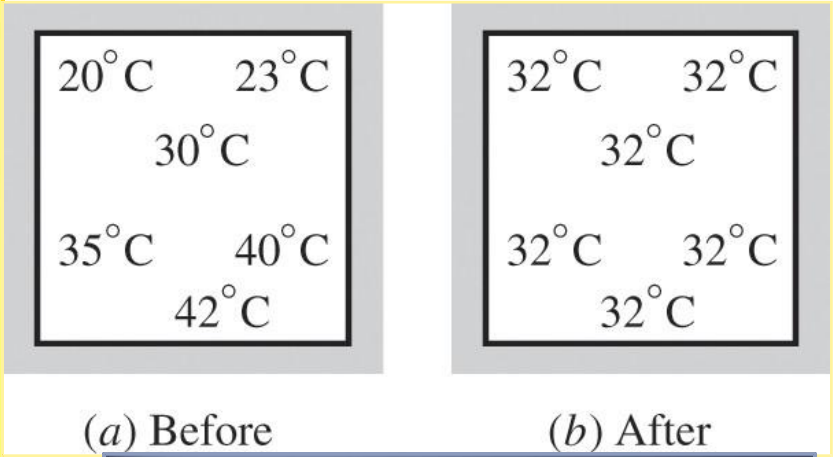
State and Equilibrium

Thermodynamics deals with ***equilibrium*** states.

- **Equilibrium**: A state of balance.
- In an equilibrium state there are no unbalanced potentials (or driving forces) within the system.
- **Thermal equilibrium**: If the temperature is the same throughout the entire system.
- **Mechanical equilibrium:** If there is no change in pressure at any point of the system with time.
- **Phase equilibrium:** If a system involves two phases and when the mass of each phase reaches an equilibrium level and stays there.
- **Chemical equilibrium:** If the chemical composition of a system does not change with time, that is, no chemical reactions occur.



A system at two different states.



A closed system reaching thermal equilibrium.

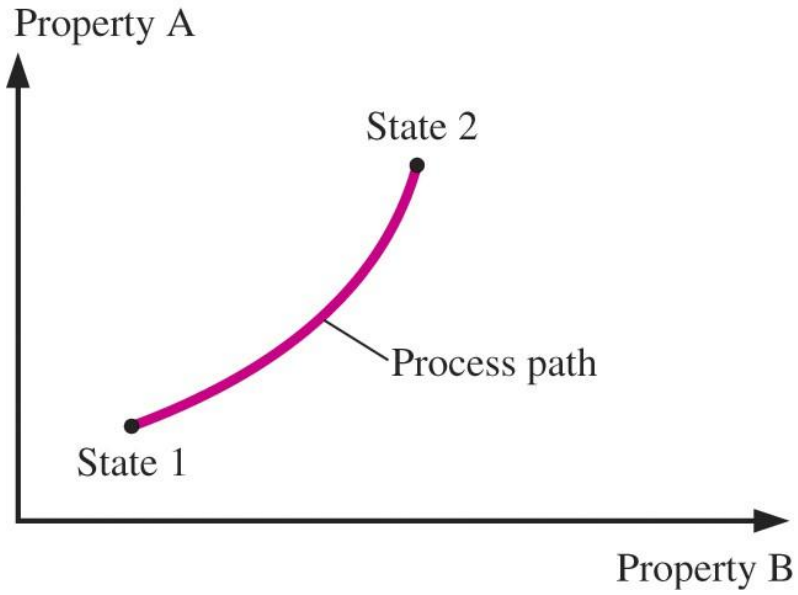
PROCESSES AND CYCLES

Process: Any change that a system undergoes from one equilibrium state to another.

Path: The series of states through which a system passes during a process.

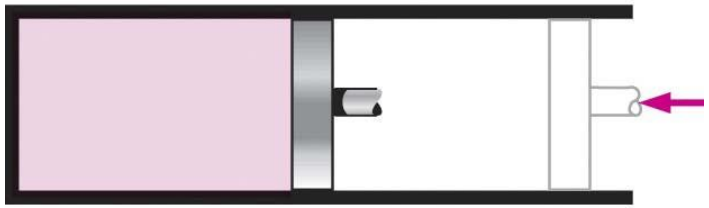
To describe a process completely, one should specify the initial and final states, as well as the path it follows, and the interactions with the surroundings.

Quasistatic or quasi-equilibrium process: When a process proceeds in such a manner that the system remains infinitesimally close to an equilibrium state at all times.





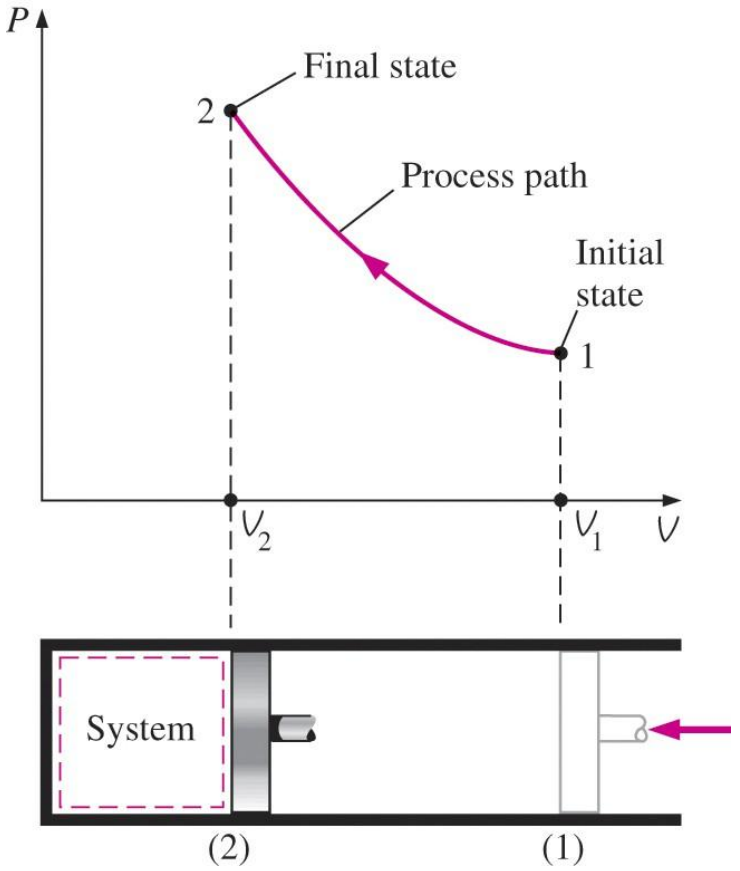
(*a*) Slow compression (quasi-equilibrium)



(*b*) Very fast compression (nonquasi-equilibrium)

PROCESSES AND CYCLES

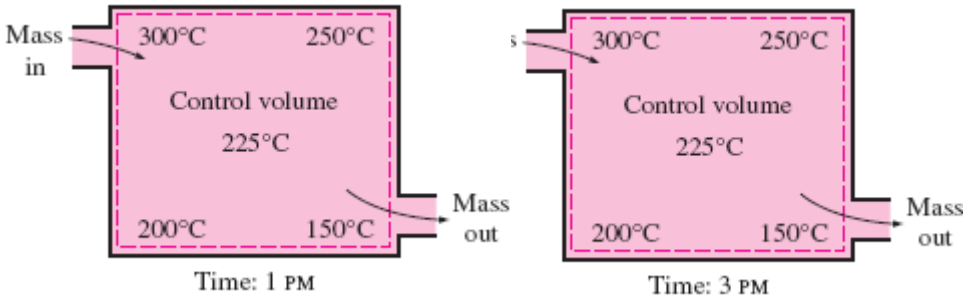
- Process diagrams plotted by employing thermodynamic properties as coordinates are very useful in visualizing the processes.
- Some common properties that are used as coordinates are temperature *T*, pressure *P*, and volume *V* (or specific volume *v*).
- The prefix *iso-* is often used to designate a process for which a particular property remains constant.
- **Isothermal process**: A process during which the temperature *T* remains constant.
- **Isobaric process**: A process during which the pressure *P* remains constant.
- **Isochoric (or isometric) process**: A process during which the specific volume *v* remains constant.
- **Cycle**: A process during which the initial and final states are identical.



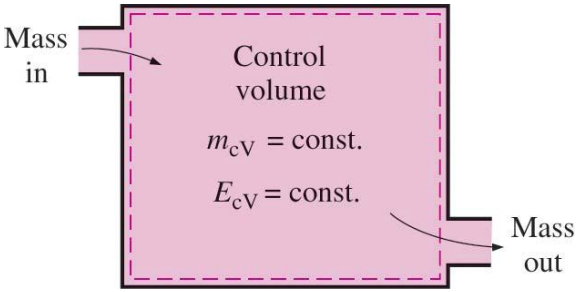
The *P-V* diagram of a compression process.

THE STEADY-FLOW PROCESS

- The term ***steady*** implies ***no change with time***. The opposite of steady is ***unsteady***, or ***transient***.
- A large number of engineering devices operate for long periods of time under the same conditions, and they are classified as ***steady-flow devices***.
- **Steady-flow process**: A process during which a fluid flows through a control volume steadily.
- Steady-flow conditions can be closely approximated by devices that are intended for continuous operation such as **turbines, pumps, boilers, condensers, and heat exchangers or power plants or refrigeration systems**.



During a steady-flow process, fluid properties within the control volume may change with position but not with time.



Under steady-flow conditions, the mass and energy contents of a control volume remain constant

ENERGY

TEMPERATURE SCALES

- All temperature scales are based on some easily reproducible states such as the freezing and boiling points of water: the *ice point* and the *steam point.*
- **Ice point**: A mixture of ice and water that is in equilibrium with air saturated with vapor at 1 atm pressure (0°C or 32°F).
- **Steam point**: A mixture of liquid water and water vapor (with no air) in equilibrium at 1 atm pressure (100°C or 212°F).
- **Celsius scale**: in SI unit system
- **Fahrenheit scale**: in English unit system
- **Thermodynamic temperature scale**: A temperature scale that is independent of the properties of any substance.
- **Kelvin scale** (SI) **Rankine scale** (E)
- A temperature scale nearly identical to the Kelvin scale is the **ideal-gas temperature scale**. The temperatures on this scale are measured using a constant-volume gas thermometer.

FORMS OF ENERGY

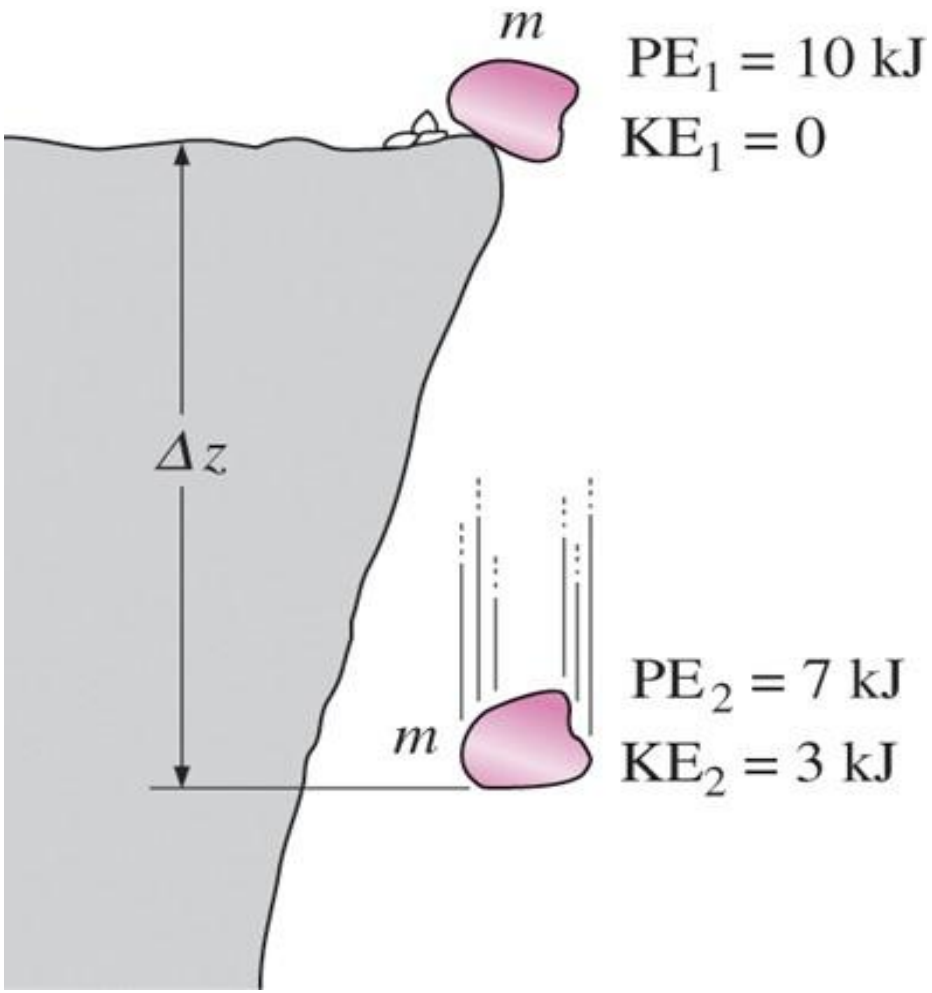
- Energy can exist in numerous forms such as thermal, mechanical, kinetic, potential, electric, magnetic, chemical, and nuclear, and their sum constitutes the **total energy, *E*** of a system.
- Thermodynamics deals only with the ***change*** of the total energy.
- **Macroscopic forms of energy**: Those a system possesses as a whole with respect to some outside reference frame, such as kinetic and potential energies.
- **Microscopic forms of energy**: Those related to the molecular structure of a system and the degree of the molecular activity.
- **Internal energy, *U***: The sum of all the microscopic forms of energy.

- **Kinetic energy, KE**: The energy that a system possesses as a result of its motion relative to some reference frame.
- **Potential energy, PE:** The energy that a system possesses as a result of its elevation in a gravitational field.



The macroscopic energy of an object changes with velocity and elevation.

Kinetic energy, KE - Potential energy, PE



$$\text{KE} = m\frac{V^2}{2} \qquad (\text{kJ})$$

Kinetic energy

$$\text{ke} = \frac{V^2}{2} \qquad (\text{kJ/kg})$$

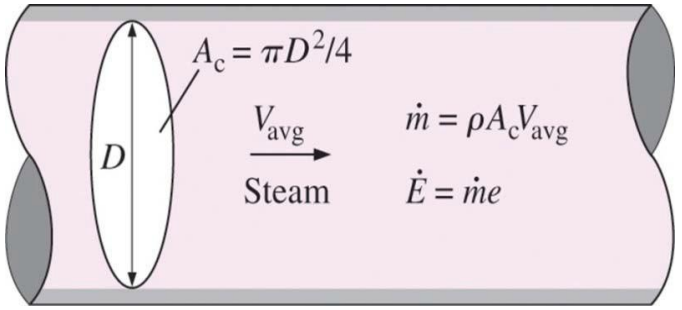
Kinetic energy per unit mass

$$\text{PE} = mgz \qquad (\text{kJ})$$

Potential energy

$$\text{pe} = gz \qquad (\text{kJ/kg})$$

Potential energy per unit mass



Mass flow rate

$$\dot{m} = \rho\dot{V} = \rho A_c V_{\text{avg}} \qquad (\text{kg/s})$$

Energy flow rate

$$\dot{E} = \dot{m}e \qquad (\text{kJ/s or kW})$$

Total energy of a system

$$E = U + \text{KE} + \text{PE} = U + m\frac{V^2}{2} + mgz \qquad (\text{kJ})$$

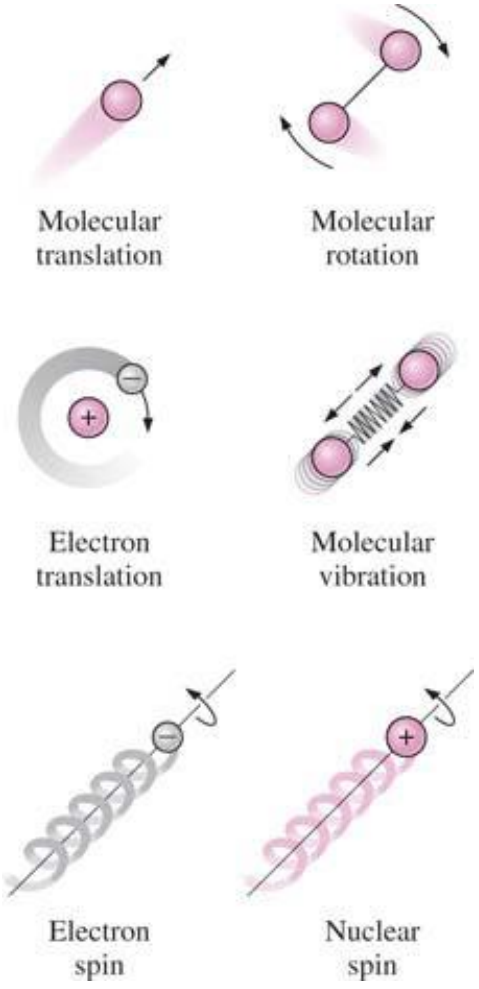
Energy of a system per unit mass

$$e = u + \text{ke} + \text{pe} = u + \frac{V^2}{2} + gz \qquad (\text{kJ/kg})$$

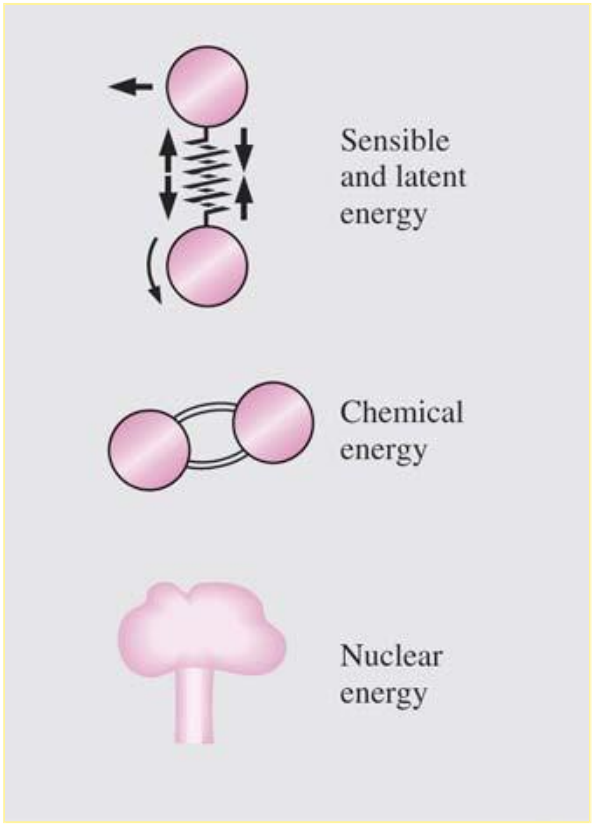
Total energy per unit mass

$$e = \frac{E}{m} \qquad (\text{kJ/kg})$$

Some Physical Insight to Internal Energy



The various forms of microscopic energies that make up *sensible* energy.



The internal energy of a system is the sum of all forms of the microscopic energies.

Sensible energy: The portion of the internal energy of a system associated with the kinetic energies of the molecules.

Latent energy: The internal energy associated with the phase of a system.

Chemical energy: The internal energy associated with the atomic bonds in a molecule.

Nuclear energy: The tremendous amount of energy associated with the strong bonds within the nucleus of the atom itself.

Thermal = Sensible + Latent

Internal = Sensible + Latent + Chemical + Nuclear

Enthalpy

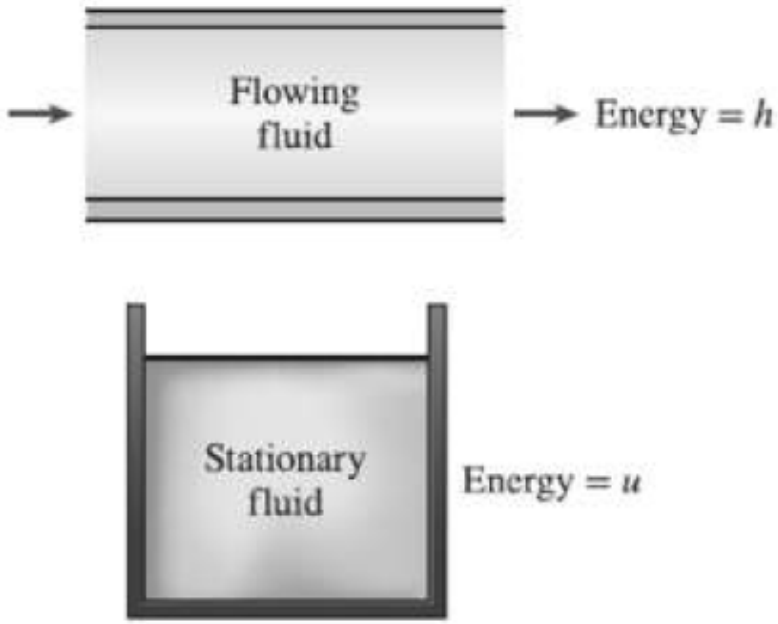
In the analysis of systems that involve fluid flow, we frequently encounter the combination of properties u and Pv.

Enthalpy: The combination $u + Pv$ is defined as enthalpy

$$h = u + Pv$$

Flow energy: The term Pv represents the flow energy of the fluid.

It is also called the flow work.



Specific Heats of Gases, Liquids, and Solids

$Pv = RT \quad P = \rho RT$ Ideal gas relation

Specific heat: The energy required to raise the temperature of a unit mass of a substance by one degree

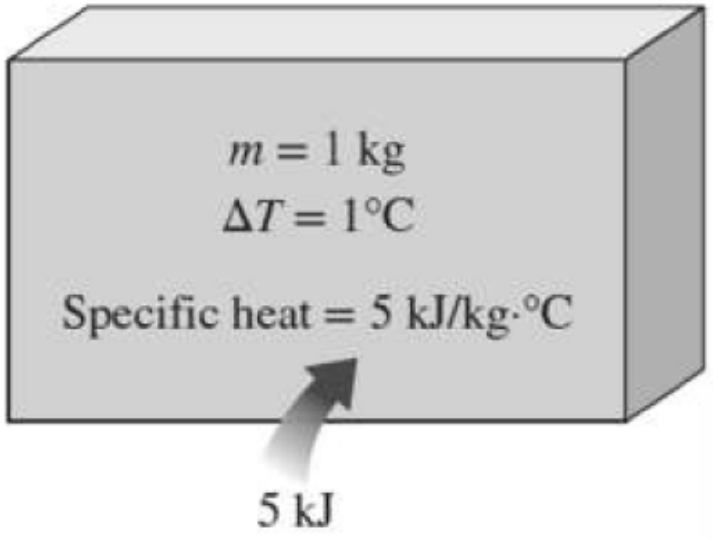
Two kinds of specific heats:

✓ specific heat at constant volume c_v

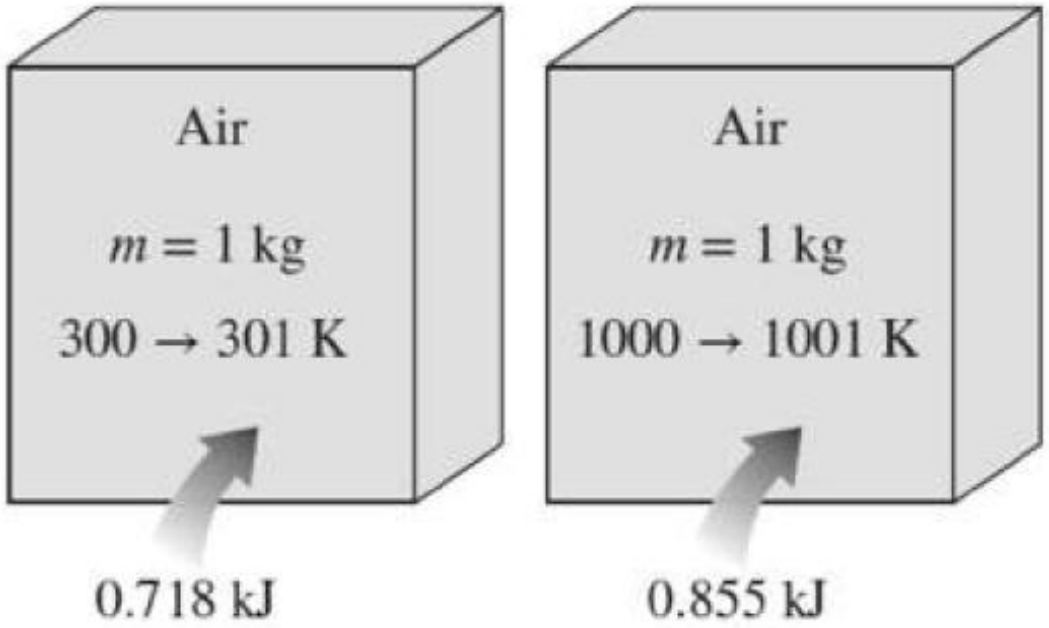
✓ specific heat at constant pressure c_p

At low pressures all real gases approach ideal gas behavior, and therefore their specific heats depend on temperature only.

$$c_p = c_v + R$$



Specific heat of a substance changes with temperature

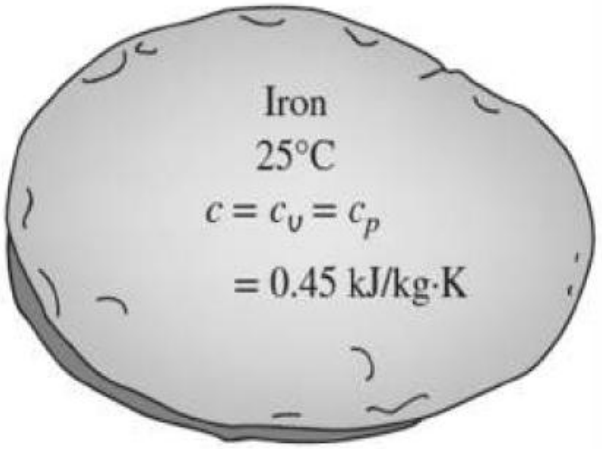


Incompressible substance

Incompressible substance: A substance whose specific volume (or density) does not change with temperature or pressure.

The constant-volume and constant-pressure specific heats are identical for incompressible substances.

The specific heats of incompressible substances depend on temperature only.



The c_v and c_p values of incompressible substances are identical and are denoted by c.


THANK YOU